Housing Market Trends and Pricing Analysis

Abstract

Real estate pricing can be affected by a variety of factors, including location, time, and property characteristics. Accurately estimating these housing prices is important for everyone, but especially for homeowners, potential buyers, policymakers, and local governments. For this project, the development team will study real estate price prediction using a large, publicly available government dataset of residential property sales in the state of Connecticut, spanning from 2001 to 2023.

The primary goal of this research is to evaluate whether modern machine learning models can capture long-term trends and nonlinear relationships in housing markets as compared to traditional approaches such as linear regression or time series analysis. Though these traditional methods are reliable for surface-level analysis and interpretability, they tend to oversimplify and struggle with identifying trends across multiple features. The team plans to explore multiple modeling techniques with increasing levels of complexity. As a baseline, a linear regression model will be used strictly for comparison. Next, tree-based models will be examined to understand nonlinear relationships between property features and sales prices. Finally, more expressive models like neural network-based approaches will be investigated, where they can provide valuable insights into different interactions between features like location, time, and property attributes.

1 Introduction

Real estate is an integral aspect of economic stability, wealth distribution, housing accessibility, and public policy. Fluctuations in the housing market not only influence individual financial decisions but also broader societal outcomes such as tax revenue, urban development, and affordability. Due to the variety of factors that go into determining sales and pricing, understanding how prices evolve over time is as much of a technical challenge as it is a problem with real consequences.

From a machine learning perspective, real estate poses an interesting modeling challenge. From long-term trends, geographic differences, and interactions between housing characteristics, Many commonly used valuation techniques rely on simplifying assumptions that make them easier to interpret but limit their ability to capture nonlinear relationships or changes in market behavior over extended periods. This creates an opportunity to evaluate whether more flexible machine learning approaches can offer meaningful improvements when it comes to predicting power and accuracy.

But apart from making predictions, this initiative is also to observe how housing markets respond to changing economic conditions over time. Through analyzing historical data, this work uncovers emerging patterns associated with market growth, stagnation, or volatility. Simultaneously, adapting algorithmic models to solve real estate problems brings up important ethical implications, shedding light on issues surrounding transparency, buyer fairness, and the potential impact on buyers and communities in the fluctuating housing market. Addressing these concerns brings attention to machine learning applications in high-stakes, safety-critical domains such as real estate.

2 The Team

Our team consists of six members, all of whom are students at Michigan State University with an interest in artificial intelligence.

Benjamin Forbes - Senior, Computer Science - forbesbe@msu.edu

Constance Chen - Senior, Computer Science- chens100@msu.edu

HyeJin (Liz) Choi - Senior, Data Science - choihye9@msu.edu

Killian Civiero - Senior, Computer Science -

civierok@msu.edu

Minh Nguyen Quang - Junior, Computer Science - nguy1290@msu.edu

Thus far, we have had two meetings, and everyone has made a conscious effort to contribute, including the division of labor for writing this report. Tasks will be handed out as they arise, but our initial labor structure is as follows:

Project Lead & Technical Writer - Benjamin Forbes

Data Engineering Lead - Constance Chen

Machine Learning Engineer - Killian Civiero

Modeling & Training Lead - Minh Nguyen Quang

Evaluation & Optimization Lead - HyeJin (Liz) Choi.

3 Related Works

Prior research has extensively explored machine learning techniques for predicting housing prices using structured socioeconomic and geographic data. A project by Nachiappan and Imran (House Price Prediction Using Machine Learning) proposes a supervised learning framework that models the relationship between dependent housing prices and independent variables such as income, property age, location, and population density. The authors primarily employ linear regression to minimize residual error between predicted and actual prices. Sharma and Harsora employ multiple machine learning models to estimate the housing prices and conclude that XGBoost is the most optimal model for predicting housing prices. Hasan and Jahan use a deep learning approach, specifically using learned embedding vectors, incorporating spatial neighborhood, raw house features, textual description, and images, in a joint embedding space to predict house prices. Almajed and Tabar analyze the bias in the housing price prediction models towards certain groups of society and evaluate several bias mitigation strategies.

From a feature perspective, several macroeconomic and geographical features act as determinants of housing prices, including economic growth, unemployment, interest rates, mortgage availability, supply constraints, and affordability ratios. Incorporating such variables can significantly improve predictive accuracy by capturing market dynamics rather than relying solely on structural housing attributes. As with all machine learning models, feature selection is a very significant predictor of the performance of the learned model.

More recent research has attempted other machine learning approaches with great success. Baldominos et al. (2018) compare linear regression with decision trees, random forests, and gradient boosting using real data containing location, property type, and time of acquisition. Their results show that nonlinear tree-based models significantly outperform linear regression. Location-related variables were identified as the most influential predictors of sale price, indicating that flexible machine learning models are better suited for capturing spatial heterogeneity in real estate markets.

4 Dataset/ Methodology

Our dataset consists of real estate sales that occurred in Connecticut from 2001 to 2023. The data was collected by a government agency and was published on data.gov. This is a very large dataset, containing 1.14 million samples with 14 features for each. From these 14, there are seven main columns that we will use to build our model: “list year”, “location”, ”town”, “address”, “assessed value”, “sale amount”, and “property type”.

Prior to model training, the raw data will undergo preprocessing to handle missing values and filter out anomalies, such as non-market transactions or extreme outliers that could skew the prediction. We will also encode categorical variables into numerical vectors and standardize continuous variables to ensure uniform scaling across features. The processed dataset will be randomly split into a training set consisting of 80% of the data and a testing set consisting of the remaining 20%.

We plan on using a linear regression model as a baseline because it can produce a real-valued prediction for a given input. We will train our model on the training data by minimizing the sum of squared errors using the least squares method. To test our model’s accuracy and generalization, we will apply our model to the testing data and evaluate performance using metrics such as the Mean Squared Error (MSE) and R-squared (R2) score.

While Linear Regression will serve as our baseline model for initial benchmarking, we intend to subsequently implement and evaluate more complex algorithms, such as Decision Trees and Neural Networks, as outlined in our abstract. This will allow us to capture non-linear patterns that a simple linear model might miss